Ayush Raj

Ghaziabad, UP — ayushcoder22@gmail.com — +91 7209768984
GitHub — LinkedIn

Education

ABES Engineering College, Ghaziabad **2023–2027**
Bachelor of Technology in Computer Science and Engineering **CGPA: 8.0/10**
Relevant Coursework: DSA, OOP, DBMS, Operating Systems, Computer Networks

Technical Skills

Languages: C, C++, Java, Python, JavaScript, SQL
Web Development: React.js, Next.js, Node.js, Express.js, Tailwind CSS, REST APIs, JWT
Databases/Tools: MongoDB, PostgreSQL, Git, GitHub, Postman, Cloudinary
Generative AI: RAG, LangChain, LangGraph, Mistral AI, ChromaDB, HuggingFace, Embeddings, Vector Databases, Agentic AI, Multi-Agent Systems
AI Tools/APIs: Tavily, BeautifulSoup, Google Gemini API, Web Search, Web Scraping

Experience

Backend Developer Intern – RoomGi **Mar 2026 – Apr 2026**

- Developed backend services using Node.js, Express.js, MongoDB, and Cloudinary for a PG Management platform.
- Designed REST APIs for authentication, property listings, bookings, and profile management using JWT.
- Optimized MongoDB queries, implemented error handling, and improved backend performance.
- Collaborated with frontend developers to integrate APIs and deliver scalable features.

Projects

Multi-Agent AI Research System GitHub

- Built a multi-agent research system using **LangGraph, LangChain, Mistral AI, Tavily, and Python** to automate complex multi-step web research.
- Designed specialized agents for **planning, web search, web scraping, information extraction, analysis, and response generation**.
- Integrated **Tavily** for real-time web search and **BeautifulSoup** for extracting relevant information from webpages.
- Used **LangGraph** to orchestrate agent workflows, maintain state, and pass structured information between specialized agents.

RAG-Based AI Knowledge Assistant GitHub

- Developed a **Retrieval-Augmented Generation (RAG)** system that answers user queries using knowledge retrieved from external documents.
- Implemented the complete pipeline including **document loading, text chunking, embeddings, vector storage, similarity search, retrieval, and generation**.
- Used **LangChain, HuggingFace embeddings, and ChromaDB** to build an efficient semantic retrieval pipeline.
- Grounded LLM responses in retrieved context to improve answer relevance and reduce unsupported or hallucinated responses.

AI Virtual Assistant GitHub Live

- Developed a **MERN-based AI Virtual Assistant** with JWT authentication, personalized user accounts, and Google Gemini API integration.
- Implemented **Speech-to-Text, Text-to-Speech, conversation history, assistant customization, and Cloudinary image uploads**.
- Built scalable **REST APIs using Node.js, Express.js, and MongoDB** for authentication, user management, and persistent conversations.
- Developed responsive React.js interfaces with protected routes and optimized frontend-backend API communication.

E-Learning Platform GitHub

- Developed a **MERN-based E-Learning platform** with JWT authentication, course management, enrollment, and progress tracking.
- Built REST APIs for **course creation, enrollment, user management, and progress tracking** using Node.js and Express.js.
- Developed responsive React.js interfaces with reusable components, course search, filtering, and modular architecture.
- Optimized MongoDB queries and API communication to improve application performance and scalability.

Achievements

- Solved **250+ DSA problems** on LeetCode.
- HackerRank Intermediate Problem Solving Certified.
- Achieved **5⋆ in C** on HackerRank.